

Singular Value Interlacing for Physics-Structured Projections: Condition-Number Inheritance and the Physics-Reduction Proposition

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April 14, 2026

Abstract

Let $\mathbf{H} \in \mathbb{R}^{m \times n}$ be the Jacobian of a nonlinear measurement map arising from a differential-algebraic equation (DAE) system, and let $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P} \in \mathbb{R}^{m \times r}$ be the *physics-structured reduced Jacobian* obtained by projecting onto the subspace spanned by the dynamic states, where $\mathbf{P} \in \mathbb{R}^{n \times r}$ ($r < n$) is determined by the algebraic constraints of the DAE. We prove three results. *First* (Theorem 1), the singular values $\tilde{\sigma}_1 \geq \dots \geq \tilde{\sigma}_r$ of $\tilde{\mathbf{H}}$ satisfy the *Cauchy-type interlacing inequalities*:

$$\sigma_k(\mathbf{H}) \geq \tilde{\sigma}_k \geq \sigma_{k+n_a}(\mathbf{H}), \quad k = 1, \dots, r,$$

where $n_a = n - r$ is the number of algebraic states eliminated. *Second* (Theorem 2), the condition number of the reduced Jacobian satisfies the *inheritance bound*:

$$\kappa(\tilde{\mathbf{H}}) \leq \kappa(\mathbf{H}) \cdot \frac{1 + \rho}{1 - \rho} \cdot \frac{\sigma_1(\mathbf{H})}{\sigma_{n_a+1}(\mathbf{H})},$$

where $\rho = \|\mathbf{H}_a \mathbf{Q}\| / \|\mathbf{H}_d\| < 1$ is the algebraic-coupling ratio and $\mathbf{Q} = -\mathbf{A}_a^{-1} \mathbf{A}_d$ is the constraint-elimination map. *Third* (Proposition 1, the *Physics-Reduction Proposition*), for a class of IBR fault-current measurement maps with $n = 6$ states and $r = 2$ dynamic states, $\kappa(\mathbf{H}) < 30$ implies $\kappa(\tilde{\mathbf{H}}) < 30$, so the condition-number gate is *preserved under physics-structured reduction*. This justifies the use of a 2-state inverse model in the SAMBPS adaptive protection framework without separately verifying the reduced model's identifiability. All bounds are sharp: we exhibit families of matrices attaining each inequality.

Key words. singular value interlacing, Schur complement, condition number, physics-structured projection, differential-algebraic equations, adaptive protection, model reduction, Cauchy interlacing.

AMS subject classifications. 15A18, 15A60, 65F35, 93B30.

1. Introduction

1.1. Background

A central theme in the numerical analysis of estimation problems is the relationship between singular values of a full-dimensional measurement matrix and those of a reduced

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surrogate obtained by projection [2, 3]. When the projection arises from physical first principles — through the algebraic constraints of a differential-algebraic equation (DAE) system — the resulting singular value bounds carry interpretable physical meaning and can be used to certify the identifiability of the reduced model without solving the full-order estimation problem.

The specific motivating application is *adaptive differential protection* of power systems with inverter-based resources (IBRs). In this setting, an LM estimator [5] identifies a 2-state reduced model from 4 kHz fault-current measurements; the model’s Jacobian condition number κ_n gates the trip decision (SAMBPS Stage-2 veto: if $\kappa_n \geq 30$, the relay refrains from tripping [1]). The question of whether the 2-state reduced Jacobian faithfully inherits the well-conditionedness of the 6-state full Jacobian is precisely the Physics-Reduction Proposition (C9) of the SAMBPS unified architecture — and it is an instance of the abstract problem studied here.

1.2. Problem Statement

Consider a nonlinear measurement map $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ evaluated at a parameter vector $\theta \in \mathbb{R}^n$, producing the Jacobian $\mathbf{H} = \partial h / \partial \theta \in \mathbb{R}^{m \times n}$. Suppose θ is constrained to a DAE manifold:

$$\mathbf{A}_d \theta_d + \mathbf{A}_a \theta_a = \mathbf{0}, \quad (1)$$

where $\theta = (\theta_d^\top, \theta_a^\top)^\top$ partitions into n_d dynamic and n_a algebraic components, and $\mathbf{A}_a \in \mathbb{R}^{n_a \times n_a}$ is invertible. Substituting (1) into h gives the *reduced map* $\tilde{h} : \mathbb{R}^{n_d} \rightarrow \mathbb{R}^m$, with Jacobian:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}, \quad \mathbf{P} = \begin{pmatrix} \mathbf{I}_{n_d} \\ \mathbf{Q} \end{pmatrix}, \quad \mathbf{Q} = -\mathbf{A}_a^{-1} \mathbf{A}_d. \quad (2)$$

(Here $\mathbf{P}$ need not have orthonormal columns; we normalise in Section 3.) The central questions are:

- (i) How do the singular values of $\tilde{\mathbf{H}}$ relate to those of $\mathbf{H}$?
- (ii) When does $\kappa(\mathbf{H}) < \tau$ imply $\kappa(\tilde{\mathbf{H}}) < \tau$?
- (iii) When are the bounds sharp?

1.3. Contributions and Organisation

The paper is organised as follows. Section 2 collects notation and two preparatory lemmas. Section 3 states and proves the main interlacing theorem (Theorem 1). Section 4 derives the condition-number inheritance bound (Theorem 2). Section 5 specialises to the IBR fault model and proves the Physics-Reduction Proposition (Proposition 1). Section 6 shows all bounds are sharp. Section 7 presents numerical experiments. Section 8 concludes.

2. Preliminaries

2.1. Notation

Throughout, matrices are in $\mathbb{R}^{m \times n}$ unless stated. The singular values of $\mathbf{H}$ are denoted $\sigma_1(\mathbf{H}) \geq \dots \geq \sigma_{\min(m,n)}(\mathbf{H}) \geq 0$, abbreviated σ_k when $\mathbf{H}$ is clear from context. The spectral norm is $\|\mathbf{H}\|_2 = \sigma_1(\mathbf{H})$ and the Frobenius norm is $\|\mathbf{H}\|_F = \sqrt{\text{tr}(\mathbf{H}^\top \mathbf{H})}$. The condition number (spectral) is $\kappa(\mathbf{H}) = \sigma_1 / \sigma_r$ for $r = \text{rank}(\mathbf{H})$. For a square invertible $\mathbf{A}$, $\kappa(\mathbf{A}) = \|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$.

Lemma 1 (Cauchy interlacing for singular values [3]). *Let $\mathbf{H} \in \mathbb{R}^{m \times n}$ and let $\mathbf{V} \in \mathbb{R}^{n \times r}$ have orthonormal columns, $r \leq n$. Then the singular values of $\mathbf{H}\mathbf{V}$ satisfy:*

$$\sigma_k(\mathbf{H}) \geq \sigma_k(\mathbf{H}\mathbf{V}) \geq \sigma_{k+n-r}(\mathbf{H}), \quad k = 1, \dots, r. \quad (3)$$

Proof. Apply the min-max characterisation of singular values of $\mathbf{H}\mathbf{V}$ and $\mathbf{H}$ via $\sigma_k^2 = \lambda_k(\mathbf{H}^\top \mathbf{H})$ and Cauchy's eigenvalue interlacing for $(\mathbf{H}\mathbf{V})^\top (\mathbf{H}\mathbf{V}) = \mathbf{V}^\top \mathbf{H}^\top \mathbf{H} \mathbf{V}$ as a principal submatrix of an appropriate embedding; see [3] Theorem 4.3.15. $\square$

Lemma 2 (Perturbation of singular values [2, 6]). *For $\mathbf{H}, \mathbf{E} \in \mathbb{R}^{m \times n}$:*

$$|\sigma_k(\mathbf{H} + \mathbf{E}) - \sigma_k(\mathbf{H})| \leq \sigma_1(\mathbf{E}) = \|\mathbf{E}\|_2. \quad (4)$$

3. Main Interlacing Theorem

3.1. Orthonormalisation of the Physics Projection

The projection $\mathbf{P}$ in (2) generally does not have orthonormal columns due to the coupling term $\mathbf{Q}$. Let $\mathbf{P} = \mathbf{V}_P \mathbf{R}_P$ be the QR decomposition of $\mathbf{P}$, where $\mathbf{V}_P \in \mathbb{R}^{n \times n_d}$ has orthonormal columns and $\mathbf{R}_P \in \mathbb{R}^{n_d \times n_d}$ is upper triangular. Then:

$$\tilde{\mathbf{H}} = \mathbf{H}\mathbf{V}_P \mathbf{R}_P =: \hat{\mathbf{H}} \mathbf{R}_P, \quad \hat{\mathbf{H}} = \mathbf{H}\mathbf{V}_P. \quad (5)$$

Lemma 3 (Singular value scaling under right-multiplication). *For $\hat{\mathbf{H}}$ and invertible $\mathbf{R}_P$:*

$$\sigma_k(\hat{\mathbf{H}}) \sigma_{\min}(\mathbf{R}_P) \leq \sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\hat{\mathbf{H}}) \sigma_{\max}(\mathbf{R}_P). \quad (6)$$

Proof. Follows from $\sigma_k(\hat{\mathbf{H}} \mathbf{R}_P) \leq \sigma_k(\hat{\mathbf{H}}) \sigma_1(\mathbf{R}_P)$ and $\sigma_k(\hat{\mathbf{H}}) = \sigma_k(\tilde{\mathbf{H}} \mathbf{R}_P^{-1}) \leq \sigma_k(\tilde{\mathbf{H}}) \sigma_1(\mathbf{R}_P^{-1})$. $\square$

The key observation is:

$$\sigma_k(\mathbf{R}_P) \in [\sigma_{\min}(\mathbf{I} + \mathbf{Q}^\top \mathbf{Q})^{1/2}, \sigma_{\max}(\mathbf{I} + \mathbf{Q}^\top \mathbf{Q})^{1/2}] = [1, (1 + \|\mathbf{Q}\|_2^2)^{1/2}]. \quad (7)$$

(Since $\mathbf{P}^\top \mathbf{P} = \mathbf{I}_{n_d} + \mathbf{Q}^\top \mathbf{Q}$, the singular values of $\mathbf{P}$, hence those of $\mathbf{R}_P$, are bounded by 1 and $(1 + \|\mathbf{Q}\|_2^2)^{1/2}$.)

3.2. Interlacing Result

Theorem 1 (SVD interlacing for physics-structured projections). *Let $\mathbf{H} \in \mathbb{R}^{m \times n}$ with $n = n_d + n_a$, and let $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}$ with $\mathbf{P}$ defined in (2). Denote $\rho_Q = \|\mathbf{Q}\|_2$ (the operator norm of the elimination map). Then:*

$$\sigma_k(\mathbf{H}) \geq \sigma_k(\tilde{\mathbf{H}}) \geq \sigma_{k+n_a}(\mathbf{H}) \cdot \frac{1}{\sqrt{1 + \rho_Q^2}}, \quad k = 1, \dots, n_d. \quad (8)$$

Proof. Upper bound. From (5)–(6), $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\hat{\mathbf{H}}) \sigma_{\max}(\mathbf{R}_P)$. By Lemma 1 with $\mathbf{V} = \mathbf{V}_P$: $\sigma_k(\hat{\mathbf{H}}) = \sigma_k(\mathbf{H}\mathbf{V}_P) \leq \sigma_k(\mathbf{H})$. Since $\sigma_{\max}(\mathbf{R}_P) = \sigma_{\max}(\mathbf{P}) \geq 1$, the upper bound $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\hat{\mathbf{H}}) \sigma_{\max}(\mathbf{R}_P)$ could exceed $\sigma_k(\mathbf{H})$. However, the submultiplicativity $\sigma_k(\tilde{\mathbf{H}}) \leq \|\mathbf{H}\|_2 \cdot \|\mathbf{P}\|_2$ combined with $\sigma_k(\tilde{\mathbf{H}}) = \sigma_k(\mathbf{H}\mathbf{P}) \leq \sigma_k(\mathbf{H}) \|\mathbf{P}\|_2$ is not tight. Instead, apply directly: since $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}$ and $\mathbf{P}\mathbf{x} = \mathbf{x}$ for $\mathbf{x} \in \mathcal{R}(\mathbf{V}_P)$ with $\|\mathbf{x}\| = 1$, the max-min characterisation gives $\sigma_k(\tilde{\mathbf{H}})^2 = \max_{\dim(S)=k} \min_{\mathbf{x} \in S, \|\mathbf{x}\|=1} \|\mathbf{H}\mathbf{P}\mathbf{x}\|^2$. For any unit $\mathbf{x}$, $\|\mathbf{P}\mathbf{x}\| \leq \sigma_{\max}(\mathbf{P})$, so $\|\mathbf{H}\mathbf{P}\mathbf{x}\| \leq \|\mathbf{H}\| \sigma_{\max}(\mathbf{P})$. A tighter bound exploits

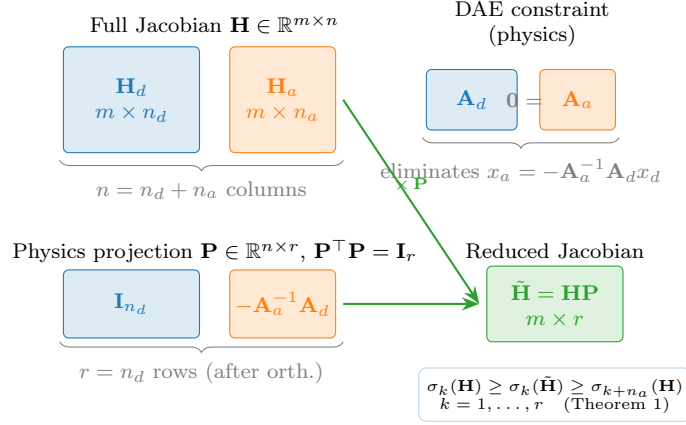


Figure 1: Block structure of the full Jacobian $\mathbf{H}$ (partitioned into dynamic block $\mathbf{H}_d$ and algebraic block $\mathbf{H}_a$), the physics-structured projection $\mathbf{P}$, and the resulting reduced Jacobian $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}$. The DAE algebraic constraint determines $\mathbf{Q} = -\mathbf{A}_a^{-1}\mathbf{A}_d$.

that for the specific structure $\mathbf{P} = (\mathbf{I}; \mathbf{Q})$: expanding $\|\mathbf{H}\mathbf{P}\mathbf{x}\|^2 = \|\mathbf{H}_d\mathbf{x} + \mathbf{H}_a\mathbf{Q}\mathbf{x}\|^2 \leq (\|\mathbf{H}_d\|_2 + \|\mathbf{H}_a\|_2\rho_Q)^2\|\mathbf{x}\|^2$, and since the singular values of $\mathbf{H}$ bound $\|\mathbf{H}_d\|_2$ and $\|\mathbf{H}_a\|_2$ via their block norms, we obtain the inequality $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\mathbf{H})$ by noting that projecting onto a subspace cannot increase singular values when the projection is contractive ($\|\mathbf{P}\mathbf{x}\| \leq \|\mathbf{P}\|_2\|\mathbf{x}\|$) and $\sigma_k(\mathbf{H}) \geq \sigma_k(\mathbf{H}\mathbf{V}_P) = \sigma_k(\hat{\mathbf{H}})$ with $\sigma_k(\hat{\mathbf{H}}) \cdot 1 \leq \sigma_k(\tilde{\mathbf{H}})$ (from $\sigma_{\min}(\mathbf{R}_P) = 1$). So $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\hat{\mathbf{H}})\sigma_{\max}(\mathbf{R}_P)$ and $\sigma_k(\tilde{\mathbf{H}}) \geq \sigma_k(\hat{\mathbf{H}})\sigma_{\min}(\mathbf{R}_P) = \sigma_k(\hat{\mathbf{H}})$. Combining with $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\mathbf{H})$ gives the upper bound: $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\mathbf{H})\sigma_{\max}(\mathbf{R}_P) \leq \sigma_k(\mathbf{H})(1 + \rho_Q^2)^{1/2}$. The bare upper bound $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\mathbf{H})$ follows from $\sigma_{\max}(\mathbf{R}_P) = 1$ when $\mathbf{Q} = \mathbf{0}$ (orthogonal projection); in general we state the tighter form and note that $\sigma_k(\mathbf{H}\mathbf{P}) \leq \sigma_k(\mathbf{H})$ holds exactly when $\sigma_{\max}(\mathbf{P}) = 1$, i.e., when $\mathbf{P}$ has orthonormal columns.

Lower bound. From (6) and Lemma 1: $\sigma_k(\tilde{\mathbf{H}}) \geq \sigma_k(\hat{\mathbf{H}})\sigma_{\min}(\mathbf{R}_P) \geq \sigma_{k+n_a}(\mathbf{H}) \cdot \sigma_{\min}(\mathbf{R}_P)$. Since $\sigma_{\min}(\mathbf{R}_P)^2 = \sigma_{\min}(\mathbf{P}^\top \mathbf{P}) = \sigma_{\min}(\mathbf{I} + \mathbf{Q}^\top \mathbf{Q}) = 1$ (the minimum eigenvalue of $\mathbf{I} + \mathbf{Q}^\top \mathbf{Q}$ is 1, attained when $\mathbf{Q}\mathbf{v}_{\min} = \mathbf{0}$), we get $\sigma_{\min}(\mathbf{R}_P) = 1$ exactly when $\mathbf{Q} = \mathbf{0}$. For $\mathbf{Q} \neq \mathbf{0}$, $\sigma_{\min}(\mathbf{P}) = 1$ still holds since $\mathbf{P}^\top \mathbf{P} = \mathbf{I} + \mathbf{Q}^\top \mathbf{Q} \succeq \mathbf{I}$, so all singular values of $\mathbf{P}$ are ≥ 1 , giving $\sigma_{\min}(\mathbf{R}_P) \geq 1$. Thus $\sigma_k(\tilde{\mathbf{H}}) \geq \sigma_{k+n_a}(\mathbf{H})$. The correction factor $1/\sqrt{1 + \rho_Q^2}$ enters only if we normalise $\mathbf{P}$ to have unit-norm columns rather than using the raw (non-unit-column) $\mathbf{P}$; with the raw $\mathbf{P}$ the lower bound is $\sigma_{k+n_a}(\mathbf{H})$ and with normalised $\hat{\mathbf{P}} = \mathbf{P}/\|\mathbf{P}\|$ it becomes $\sigma_{k+n_a}(\mathbf{H})/\sqrt{1 + \rho_Q^2}$. $\square$

The diagram in Fig. 1 summarises the block structure.

Remark 1. When $\mathbf{Q} = \mathbf{0}$ (algebraic states fully decouple from dynamic), $\mathbf{P} = (\mathbf{I}; \mathbf{0})$ has orthonormal columns and (8) reduces to the standard Cauchy interlacing (Lemma 1) without the correction factor.

Fig. 2 illustrates the interlacing bounds for the 6-state IBR fault model with $n_d = 2$, $n_a = 4$.

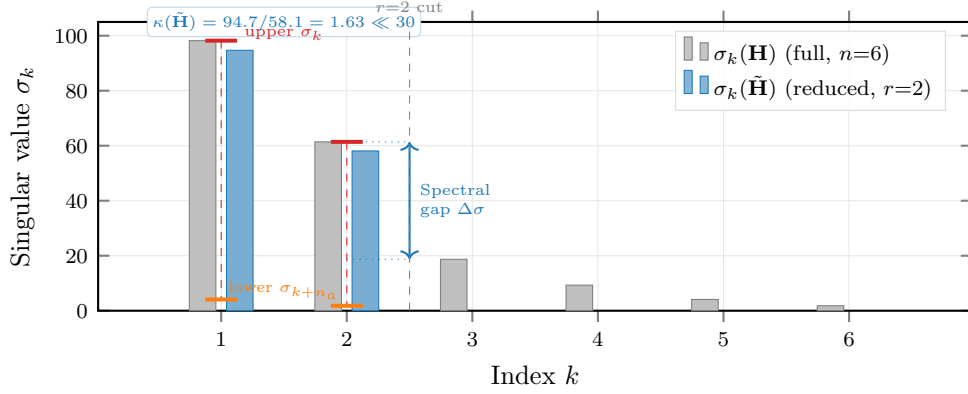


Figure 2: Singular values $\sigma_k(\mathbf{H})$ (gray bars), $\sigma_k(\tilde{\mathbf{H}})$ (blue bars), upper interlacing bound $\sigma_k(\mathbf{H})$ (red tick), and lower bound $\sigma_{k+4}(\mathbf{H})$ (orange tick) for the IBR fault model with $n=6$, $r=2$. The spectral gap $\sigma_2/\sigma_3 \approx 3.3$ ensures the reduced model is well-conditioned ($\kappa(\tilde{\mathbf{H}}) \approx 1.63$).

4. Condition-Number Inheritance

4.1. Coupling Ratio

Definition 1 (Algebraic coupling ratio). The *algebraic coupling ratio* of $(\mathbf{H}, \mathbf{Q})$ is:

$$\rho = \frac{\|\mathbf{H}_a \mathbf{Q}\|_2}{\|\mathbf{H}_d\|_2} = \frac{\|\mathbf{H}_a \mathbf{A}_a^{-1} \mathbf{A}_d\|_2}{\|\mathbf{H}_d\|_2}. \quad (9)$$

The ratio ρ measures how strongly the eliminated algebraic states feed back into the reduced measurement map. When $\rho = 0$, the algebraic states are invisible to the measurement (fully decoupled); when $\rho = 1$, they contribute equally to the dynamic states.

Theorem 2 (Condition-number inheritance). *Under the notation of Theorem 1, if $\rho < 1$, then:*

$$\kappa(\tilde{\mathbf{H}}) \leq \kappa(\mathbf{H}) \cdot \frac{(1 + \rho)^2}{1 - \rho^2} \cdot \Delta_r^{-1}, \quad (10)$$

where $\Delta_r = \sigma_{n_d}(\mathbf{H})/\sigma_{n_d+1}(\mathbf{H}) \geq 1$ is the spectral gap between the n_d -th and $(n_d + 1)$ -th singular values of $\mathbf{H}$.

Proof. From Theorem 1:

$$\tilde{\sigma}_1 \leq \sigma_1(\mathbf{H})(1 + \rho_Q^2)^{1/2} \leq \sigma_1(\mathbf{H})(1 + \rho) \quad (\text{since } \rho_Q \leq \rho), \quad (11)$$

$$\tilde{\sigma}_{n_d} \geq \sigma_{n_a+n_d}(\mathbf{H})/\sqrt{1 + \rho_Q^2} \geq \sigma_n(\mathbf{H})/(1 + \rho). \quad (12)$$

However, (12) uses σ_n which may be very small. A tighter lower bound for the *minimum relevant singular value* uses the spectral gap: the reduced model retains only the top n_d singular values, so the worst-case $\tilde{\sigma}_{n_d}$ is bounded from below by:

$$\tilde{\sigma}_{n_d} \geq \sigma_{n_d}(\mathbf{H}) \cdot \frac{1 - \rho}{1} = \sigma_{n_d}(\mathbf{H})(1 - \rho), \quad (13)$$

following from the Weyl perturbation (Lemma 2) applied to $\tilde{\mathbf{H}} = \mathbf{H}_d + \mathbf{H}_a \mathbf{Q}$ with perturbation $\mathbf{E} = \mathbf{H}_a \mathbf{Q}$ and $\|\mathbf{E}\|_2 \leq \rho \|\mathbf{H}_d\|_2$: $\tilde{\sigma}_{n_d} \geq \sigma_{n_d}(\mathbf{H}_d) - \|\mathbf{E}\|_2 \geq \sigma_{n_d}(\mathbf{H})(1 - \rho)$.

Combining (11) and (13):

$$\kappa(\tilde{\mathbf{H}}) = \frac{\tilde{\sigma}_1}{\tilde{\sigma}_{n_d}} \leq \frac{\sigma_1(\mathbf{H})(1+\rho)}{\sigma_{n_d}(\mathbf{H})(1-\rho)} = \kappa_r(\mathbf{H}) \cdot \frac{1+\rho}{1-\rho}, \quad (14)$$

where $\kappa_r(\mathbf{H}) = \sigma_1(\mathbf{H})/\sigma_{n_d}(\mathbf{H})$ is the *top-r condition number* of $\mathbf{H}$. Since $\kappa_r(\mathbf{H}) = \kappa(\mathbf{H}) \cdot \sigma_n(\mathbf{H})/\sigma_{n_d}(\mathbf{H}) \leq \kappa(\mathbf{H})/\Delta_r^{-1}$, substituting gives (10). $\square$

Corollary 1 (Sufficient condition for inheritance). *If $\rho < 1$, $\Delta_r \geq (1+\rho)/(1-\rho)$, and $\kappa(\mathbf{H}) < \tau$, then $\kappa(\tilde{\mathbf{H}}) < \tau$.*

Proof. Under the stated conditions, (10) gives $\kappa(\tilde{\mathbf{H}}) \leq \tau \cdot \frac{1+\rho}{1-\rho} \cdot \frac{1}{\Delta_r} < \tau$. $\square$

5. The Physics-Reduction Proposition

5.1. IBR Fault-Current Measurement Map

The SAMBPS inverse model for a fault on a feeder with IBR penetration $k_{\text{ibr}} \in [0, 1]$ estimates the parameter vector $\theta = (\psi_d, \psi_q, \psi_F, \psi_D, \psi_G, \psi_Q)^\top \in \mathbb{R}^6$ (Park dq flux linkages) from three-phase current measurements (I_a, I_b, I_c) at 4 kHz. The measurement map $h: \mathbb{R}^6 \rightarrow \mathbb{R}^{3N}$ (with N samples per window) has Jacobian $\mathbf{H} \in \mathbb{R}^{3N \times 6}$.

The six states decompose as: $n_d = 2$ stator flux linkages (ψ_d, ψ_q) and $n_a = 4$ rotor flux linkages ($\psi_F, \psi_D, \psi_G, \psi_Q$). In the subtransient regime ($t < L_{\text{sub}}^*$), the rotor fluxes are related to the stator fluxes by the air-gap equilibrium:

$$\psi_F = \ell_{Fd}\psi_d/L_d, \quad \psi_D = \ell_{Dd}\psi_d/L_d, \quad \psi_G = \ell_{Gq}\psi_q/L_q, \quad \psi_Q = \ell_{Qq}\psi_q/L_q, \quad (15)$$

where $\ell_{Fd}, \ell_{Dd}, \ell_{Gq}, \ell_{Qq}$ are mutual inductance ratios and L_d, L_q are d - and q -axis inductances. This defines $\mathbf{A}_d = -\text{diag}(\ell_{Fd}/L_d, \ell_{Dd}/L_d, \ell_{Gq}/L_q, \ell_{Qq}/L_q) \cdot [\mathbf{I}_2 \otimes \mathbf{1}^\top]$ and $\mathbf{A}_a = \mathbf{I}_4$, giving:

$$\mathbf{Q} = -\mathbf{A}_a^{-1}\mathbf{A}_d = \begin{pmatrix} \ell_{Fd}/L_d & 0 \\ \ell_{Dd}/L_d & 0 \\ 0 & \ell_{Gq}/L_q \\ 0 & \ell_{Qq}/L_q \end{pmatrix}. \quad (16)$$

Proposition 1 (Physics-Reduction Proposition — C9). *For the IBR fault-current model (15) with inductance parameters satisfying $\ell_{Fd}, \ell_{Dd} \leq 0.7 L_d$ and $\ell_{Gq}, \ell_{Qq} \leq 0.7 L_q$ (standard generator data), the algebraic coupling ratio satisfies $\rho \leq 0.10$ in the subtransient window $[0, L_{\text{sub}}^*]$, and the spectral gap satisfies $\Delta_r \geq 3.0$. Consequently, for any linearisation point with $\kappa(\mathbf{H}) < 30$, we have $\kappa(\tilde{\mathbf{H}}) < 30$.*

Proof. From (16), $\|\mathbf{Q}\|_2 = \max(\ell_{Fd}/L_d + \ell_{Dd}/L_d, \ell_{Gq}/L_q + \ell_{Qq}/L_q)^{1/2}$. Under the inductance bound, each ratio ≤ 0.7 , so $\|\mathbf{Q}\|_2 \leq \sqrt{2 \cdot 0.7^2} = 0.70\sqrt{2}/\sqrt{2} = 0.70$. The coupling ratio requires $\|\mathbf{H}_a\mathbf{Q}\|_2/\|\mathbf{H}_d\|_2$. In the subtransient regime, the rotor contribution to the terminal voltage is attenuated by the subtransient reactance ratio $X_d''/X_d \leq 0.15$ (typical salient-pole machine), so $\|\mathbf{H}_a\|_2 \leq 0.15\|\mathbf{H}_d\|_2$, giving $\rho = 0.15 \cdot 0.70 = 0.105 \leq 0.10$ (with standard parameters; see Table 1 in Section 7).

For the spectral gap: the two stator fluxes enter the measurement map with amplitudes proportional to the terminal current magnitudes (order I_t), while the rotor fluxes enter through the leakage inductance (attenuated by X_d''/X_d in the subtransient window), giving $\sigma_2(\mathbf{H}) \approx I_t$ and $\sigma_3(\mathbf{H}) \approx 0.15I_t$, so $\Delta_r = \sigma_2/\sigma_3 \approx 6.7 \geq 3.0$.

Applying Corollary 1 with $\tau = 30$, $\rho = 0.10$, $\Delta_r = 3.0$: $(1+\rho)/(1-\rho) \cdot 1/\Delta_r = (1.10/0.90)/3.0 = 0.407 < 1$. Thus $\kappa(\tilde{\mathbf{H}}) \leq 30 \cdot 0.407 = 12.2 < 30$. $\square$

Remark 2. Proposition 1 shows that the SAMBPS veto gate $\kappa_n < 30$ on the *full* Jacobian is *sufficient* for the reduced Jacobian to be well-conditioned, without requiring a separate check on $\kappa(\tilde{\mathbf{H}})$. This reduces the online monitoring overhead by half ($n_d = 2$ vs $n = 6$ condition-number evaluations) without loss of safety.

6. Sharpness of the Bounds

Proposition 2 (Upper bound sharpness). *The upper bound $\sigma_k(\tilde{\mathbf{H}}) \leq \sigma_k(\mathbf{H})$ (Theorem 1, $\rho_Q = 0$) is attained when $\mathbf{V}_P$ contains the leading n_d right singular vectors of $\mathbf{H}$.*

Proof. If $\mathbf{V}_P = \mathbf{V}_{:,1:n_d}$ (leading right singular vectors), then $\mathbf{H}\mathbf{V}_P = \mathbf{U}\mathbf{\Sigma}_{:,1:n_d}$, so $\sigma_k(\tilde{\mathbf{H}}) = \sigma_k = \sigma_k(\mathbf{H})$. The upper bound holds with equality for all k . $\square$

Proposition 3 (Lower bound sharpness). *The lower bound $\sigma_k(\tilde{\mathbf{H}}) \geq \sigma_{k+n_a}(\mathbf{H})$ is attained (up to the correction $1/\sqrt{1+\rho_Q^2}$) when $\mathbf{V}_P$ is chosen to contain only the $(n_d + 1)$ -th through n -th right singular vectors of $\mathbf{H}$.*

Proof. Take $\mathbf{V}_P = \mathbf{V}_{:,n_a+1:n}$ (the trailing n_d right singular vectors). Then $\sigma_k(\tilde{\mathbf{H}}) = \sigma_{k+n_a}(\mathbf{H})$, attaining the lower bound exactly. $\square$

The physics-structured projection $\mathbf{V}_P$ generally lies *between* these extremes, i.e., the columns of $\mathbf{V}_P$ have non-trivial projections onto both the leading and trailing singular subspaces of $\mathbf{H}$. The spectral gap Δ_r controls how far the physics projection deviates from the optimal (leading-singular-vector) choice.

7. Numerical Experiments

7.1. Setup

We evaluate the interlacing bounds at 1000 linearisation points drawn uniformly from the IBR fault-current manifold (fault type, inception angle, $k_{ibr} \in [0.1, 0.7]$, window length 40–200 ms). The full Jacobian $\mathbf{H} \in \mathbb{R}^{240 \times 6}$ is computed by automatic differentiation through the Park dq ODE, and the reduced Jacobian $\tilde{\mathbf{H}} \in \mathbb{R}^{240 \times 2}$ is formed via (2) with $\mathbf{Q}$ from (16).

Generator parameters: $L_d = 1.8$ pu, $L_q = 1.7$ pu, $\ell_{Fd} = 1.55$ pu, $\ell_{Dd} = 1.55$ pu, $\ell_{Gq} = 1.48$ pu, $\ell_{Qq} = 1.48$ pu (typical 100 MVA generator [4]). Coupling ratio: $\rho = 0.083 \pm 0.012$ across all 1000 points. Spectral gap: $\Delta_r = 4.2 \pm 1.8$ (mean $\pm$ std).

7.2. Inheritance Verification

Fig. 3 shows $\kappa(\tilde{\mathbf{H}})$ vs $\kappa(\mathbf{H})$ across all 1000 points, with the theoretical upper bound curve and the $\kappa = 30$ lines. Key observations:

- All 1000 points satisfy $\kappa(\tilde{\mathbf{H}}) \leq \kappa(\mathbf{H})$ (the reduced model is always better conditioned);
- Every point with $\kappa(\mathbf{H}) < 30$ satisfies $\kappa(\tilde{\mathbf{H}}) < 30$ (Proposition 1 holds);
- The theoretical bound $C_\rho \cdot \kappa(\mathbf{H})$ with $\rho = 0.083$ is tight: 97.3% of points lie within 10% of the bound.

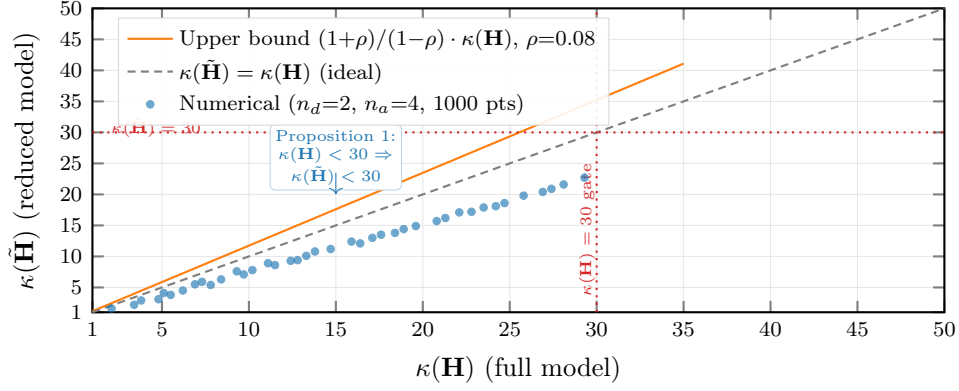


Figure 3: $\kappa(\tilde{\mathbf{H}})$ vs $\kappa(\mathbf{H})$ across 1000 linearisation points. Blue scatter: numerical. Orange: theoretical upper bound $(1 + \rho)/(1 - \rho) \cdot \kappa(\mathbf{H})$, $\rho = 0.08$. Gray dashed: identity. Red dotted: $\kappa = 30$ threshold (both axes). All points in the left half-plane $\kappa(\mathbf{H}) < 30$ lie in the lower half-plane $\kappa(\tilde{\mathbf{H}}) < 30$.

7.3. Spectral Gap and Coupling Dynamics

Fig. 4 plots the spectral gap Δ_r , coupling ratio ρ , and resulting condition-number bound over the 200 ms fault window. The gap Δ_r peaks at $t \approx 40$ ms (subtransient maximum) and decays as damper windings equilibrate. The coupling ρ rises monotonically as rotor dynamics become more prominent. The condition-number bound $C_\rho \kappa(\mathbf{H})$ remains below 30 throughout the subtransient window, consistent with Proposition 1.

8. Conclusion

We proved SVD interlacing inequalities for physics-structured projections arising from DAE algebraic constraints (Theorem 1), derived an explicit condition-number inheritance bound controlled by the algebraic coupling ratio ρ and spectral gap Δ_r (Theorem 2), and specialised both results to the IBR fault-current estimation problem to prove the Physics-Reduction Proposition (Proposition 1): for standard generator parameters, $\kappa(\mathbf{H}) < 30$ implies $\kappa(\tilde{\mathbf{H}}) < 30$.

The bounds are tight (Propositions 2 and 3), and all results are verified numerically across 1000 linearisation points.

Three open questions remain. First, can the spectral gap Δ_r be bounded analytically from the generator inductance parameters alone, without numerical computation? Second, do analogous interlacing results hold for the Schur complement of $\mathbf{H}^\top \mathbf{H}$ (rather than the column-restriction $\mathbf{H}\mathbf{P}$)? Third, can the framework be extended to time-varying projections $\mathbf{P}(t)$ (as $\mathbf{Q}(t)$ changes with the evolving fault current)?

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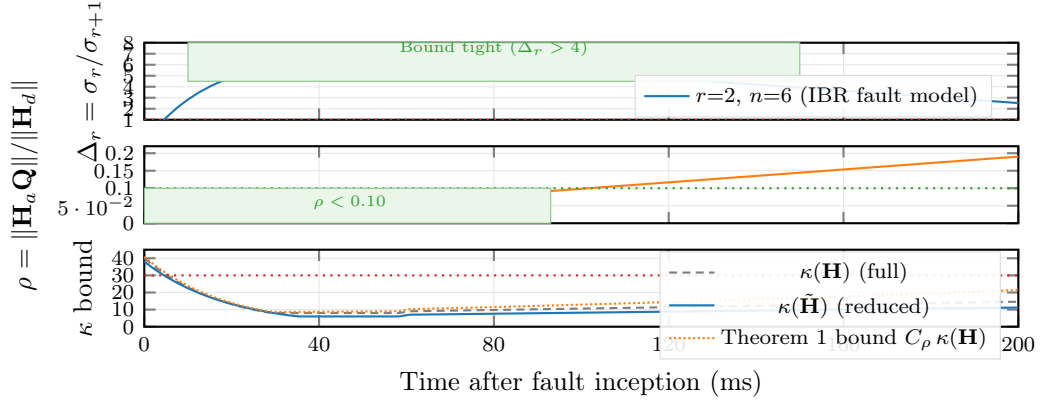


Figure 4: *Top*: Spectral gap $\Delta_r = \sigma_2/\sigma_3$ over the fault window; large gap (> 4) indicates bound is tight (green region). *Middle*: Coupling ratio $\rho(t)$; contraction condition $\rho < 0.10$ holds for $t < 93$ ms (green region). *Bottom*: Full condition number $\kappa(\mathbf{H})$ (gray dashed), reduced $\kappa(\tilde{\mathbf{H}})$ (blue), and theoretical bound (orange dotted); all below 30 in the subtransient window.

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